Unit 4: Turning Effect of Forces

Complete notes for Unit 4 of *A Textbook of Physics, Grade IX* (Khyber Pakhtunkhwa Textbook Board, Peshawar) - every definition, formula and worked example from the book, plus the full exercise solved: MCQs with an answer key, all six numericals step by step, and short answers to the conceptual questions.

> **How to use these notes.** Read one section at a time, copy each formula into your notebook, and redo every solved example yourself before looking at the answer. The last five pages are pure exam practice - the book's own exercise, solved.

What this unit covers

Book section	Topic	Book pages
4.1	Force diagrams	100
4.2	Parallel forces and addition of forces	101-103
4.3	Resolution of forces	104-107
4.4	Centre of mass and centre of gravity	108-111
4.5	Torque or moment of a force	112-114
4.6	Couple	115
4.7	Equilibrium and principle of moments	116-120
4.8	Stability	121-123
Exercise	MCQs, conceptual, comprehensive and numerical questions	126-128

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After studying this unit, you should be able to

These are the student learning outcomes printed on the first page of the unit in the textbook (page 99). Tick each one off as you master it.

- Define like and unlike parallel forces.
- State the head-to-tail rule of vector addition of forces.
- Describe how a force is resolved into its perpendicular components.
- Determine the magnitude and direction of a force from its perpendicular components.
- Define moment of force or torque as: moment = force x perpendicular distance from pivot to the line of action of force.
- Explain the turning effect of a force by relating it to everyday life.
- State the principle of moments.
- Define the centre of mass and centre of gravity of a body.
- Define a couple as a pair of forces tending to produce rotation.
- Prove that the couple has the same moment about all points.
- Define equilibrium and classify its types, quoting examples from everyday life.
- State the two conditions for equilibrium of a body.
- Solve problems on simple balanced systems when bodies are supported by one pivot only.
- Describe the states of equilibrium and classify them with common examples.
- Explain the effect of the position of the centre of mass on the stability of simple objects.

> **The big idea of the unit.** In Units 2 and 3 a force changed a body's translational motion - it moved in a straight line. In this unit the same force, applied away from a fixed point, produces **rotational motion**. The measure of that turning effect is the **torque**, and almost every topic in the unit - couple, equilibrium, stability - is built on it.

4.1 Force diagrams

To study the forces acting on an object, we draw a **force diagram**. Force is a vector, so it is represented by an arrow: the length of the arrow shows the magnitude and the arrowhead shows the direction. In a force diagram the object is reduced to a **dot at its centre**, and all the forces acting on it are drawn as arrows starting from that dot.

> **Book example (Figure 4.1).** A textbook resting on a table has two forces on it: its **weight**, pulling it downward, and the **normal force** of the table, pushing it upward. The two forces are equal and opposite, so they are **balanced forces** - they cancel each other and the book behaves as if no force acts on it. The same book in **free fall** has only its weight acting (air resistance neglected); the force is **unbalanced**, so the book accelerates downward.

Parallel forces

Forces whose directions (lines of action) are parallel to each other are called **parallel forces**, whether they point the same way or opposite ways.

> **Like parallel forces** act in the *same* direction. Example from the book: lifting a box with two supports - both supports push upward on the box, though one support may push harder than the other.

> **Unlike parallel forces** act in *opposite* directions. Example from the book: both hands turning the steering wheel of a car - one hand pushes up on one side while the other pulls down on the opposite side.



Like and unlike parallel forces, after Figures 4.2 of the textbook.

4.2 Addition of forces - the head-to-tail rule

Addition of forces means finding a single force that produces the same effect as all the given forces acting together. That single force is called the **resultant force** (or net force - the two terms mean the same thing).

Parallel forces: simple rules

- **Like parallel forces:** add the magnitudes. 5 N + 5 N = 10 N in the same direction; 5 N + 10 N = 15 N.
- **Unlike parallel forces:** subtract the magnitudes; the resultant points in the direction of the larger force. 10 N up with 5 N down gives 5 N up; 5 N against 10 N gives 5 N in the direction of the 10 N force; equal opposite forces (5 N and 5 N) give a resultant of **zero**.

Forces at an angle: head-to-tail rule

When forces are not parallel, they are added geometrically. The book gives the rule in steps:

1. Choose a suitable scale (for example, 1 cm = 5 N) and draw the axes.
2. Draw the first force vector to scale, at its given angle with the x-axis.
3. Place the **tail of the second vector on the head of the first**, keeping its own direction.
4. Join the tail of the first vector to the head of the last vector. This closing vector is the **resultant**.
5. Measure the resultant's length and multiply by the scale to get its magnitude; measure its angle with a protractor.

WORKED EXAMPLE - FROM THE BOOK (PAGE 104)

Two persons pull a cart. Their forces are drawn to the scale 1 cm = 5 N and added head to tail. The resultant vector measures about 6.2 cm, and its angle with the x-axis measures 30 degrees.

Resultant = 6.2 x 5 = **31 N, at 30 degrees with the x-axis**.

> **Order does not matter.** Drawing the second force first and then the first gives the same resultant - vector addition is commutative. The head-to-tail rule also works for more than two forces: keep joining them head to tail and close the polygon.

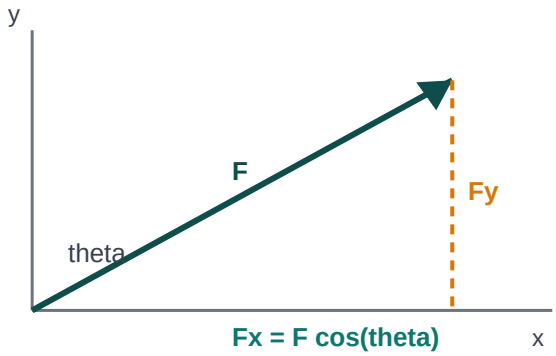
4.3 Resolution of forces

The process of splitting a force vector into two or more force vectors is called **resolution of forces**, and the parts obtained are called its **components**. Resolution is the reverse of addition. When the components are perpendicular to each other, they are called **rectangular components**. Resolving a force into rectangular components lets us add forces by calculation instead of by ruler and protractor - the analytical method is more accurate than the graphical method.

Let a force **F** act at an angle **theta** with the x-axis. Its two rectangular components are:

Fx = F cos(theta) | Fy = F sin(theta)

Fx is the horizontal component, Fy the vertical component



A force and its rectangular components (textbook Figure 4.6, redrawn).

Finding the force back from its components

By Pythagoras' theorem on the right-angled triangle of components:

F = square root of (Fx^2 + Fy^2) | theta = $\tan^{-1}(Fy / Fx)$

> **Remember:** a component can never be greater than the force itself, because cos(theta) and sin(theta) are never greater than 1. At theta = 45 degrees the two components are equal - a favourite MCQ point.

Solved examples - resolution

EXAMPLE 4.2 - PULLING A BOX ON THE FLOOR (BOOK, PAGE 106)

A force of **20 N** is applied at an angle of **60 degrees** with the horizontal. Find the horizontal and vertical components of this force.

Given: F = 20 N, theta = 60 degrees. **Required:** Fx = ?, Fy = ?

Fx = F cos(theta) = 20 x cos 60 = 20 x 0.5 = **10 N**

Fy = F sin(theta) = 20 x sin 60 = 20 x 0.866 = **17.32 N**

Only the horizontal component, 10 N, actually drags the box forward; the vertical component partly lifts it, reducing friction.

PRACTICE - IN THE STYLE OF ASSIGNMENT 4.1 (BOOK, PAGE 104)

Two forces are applied: 25 N at 20 degrees with the x-axis and 10 N at 60 degrees with the x-axis. Find the resultant analytically.

Resolve each force. Force 1: Fx = 25 cos 20 = 23.49 N; Fy = 25 sin 20 = 8.55 N. Force 2: Fx = 10 cos 60 = 5.0 N; Fy = 10 sin 60 = 8.66 N.

Total Fx = 23.49 + 5.0 = 28.49 N. Total Fy = 8.55 + 8.66 = 17.21 N.

Resultant F = square root of (28.49^2 + 17.21^2) = square root of (811.7 + 296.2) = square root of 1107.9 = **about 33.3 N**

Direction: theta = $\tan^{-1}$(17.21 / 28.49) = $\tan^{-1}$(0.604) = **about 31 degrees with the x-axis**.

The general method, worth memorising. (1) Resolve every force into Fx and Fy. (2) Add all Fx values together and all Fy values together, keeping signs. (3) Combine the two totals with Pythagoras for the magnitude and $\tan^{-1}$(Fy/Fx) for the direction. This single method solves every resultant-force question in the unit.

Rotational motion - the key terms

Rotational motion. Motion in which all points of an object move about a single fixed axis is called rotational motion - for example, the blades of a blender or a spinning satellite (textbook, page 108).

Rigid body. A body of fixed form that does not distort or deform (change shape) as it moves, so the distances between its particles stay fixed. Treating bodies as rigid makes rotational dynamics much simpler.

Axis of rotation. The line about which rotation takes place. This line stays stationary during the rotation while all other points of the body move in circles around it. It may be a pivot, a hinge, or any other support.

4.4 Centre of mass and centre of gravity

Centre of mass (CM). The point of a body about which its mass is equally distributed in all directions. The centre of mass moves exactly as a single particle would move under the same net force, even while the body itself rotates.

Centre of gravity (CG). The point through which the whole weight of the body acts, whatever the body's position. For ordinary bodies in a uniform gravitational field, the CM and the CG are at the same point. They differ only when the gravitational field is not uniform over the body (for example, a very tall object) - this is the point tested in MCQ 10 of the exercise.

Symmetric bodies of uniform density

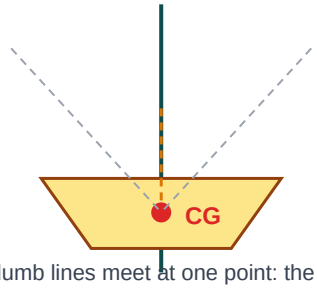
For a homogeneous sphere, cube, circular disc or rectangular plate, the CM/CG lies at the **geometric centre**. For a right circular cylinder or a cone it lies on the **axis of symmetry**; for a uniform rod it is at the **mid-point** (textbook Figure 4.9). Note that the CM/CG may even lie *outside* the material of the body - for example, at the empty centre of a ring or a bangle.

Finding the CM/CG of an irregular body

When the mass is not distributed uniformly - for example, a light wooden rod with a heavy metal piece attached - symmetry cannot locate the CM/CG. The book finds it by the **plumb-line (hanging) method** (Figure 4.10, using a bicycle):

1. Hang the body freely from one point and let it come to rest.
2. Hang a plumb line (a string with a weight) from the same point and mark the vertical line it makes across the body. The CM/CG lies somewhere on this line.
3. Repeat from a second point, and then from a third point, marking a new line each time.
4. The point where all the lines intersect is the **centre of gravity** - which, for such a body, is also its centre of mass.

For a flat irregular lamina in the laboratory, the same method is used with three holes near the edges: pin the lamina through one hole, draw the plumb line, and repeat; the three lines cross at the CG.



Three plumb lines meet at one point: the CG

The plumb-line method: every line passes through the CG.

> **Why the method works.** A freely hanging body comes to rest only when its CG lies on the vertical line through the point of support. Each hanging therefore gives one line that must contain the CG, and two or three such lines fix the point exactly.

Balance point

Support a body exactly under its CM/CG and it balances in any position - this is why the same point is also called the **balance point** of the body.

4.5 Torque or moment of a force

A force applied to a body that can turn about a fixed point produces rotation: pushing a door, tightening a nut with a spanner, turning a steering wheel. This turning effect is the torque.

> **Torque (moment of a force).** The turning effect produced in a body about a fixed point due to an applied force is called torque, or moment of force. Physicists usually say *torque*; engineers usually say *moment*. Torque does for rotation what force does for translation: it is the cause of changes in rotational motion.

Torque = Force x Moment arm - tau = F x d

tau (torque) in newton metre (Nm); F in newton (N); d in metre (m)

The **moment arm** (or lever arm) **d** is the *perpendicular* distance from the axis of rotation to the line of action of the force. Two consequences follow at once:

- Torque increases with a **larger force** and with a **longer moment arm**. That is why door handles are fitted far from the hinges, and why a long spanner loosens a rusty nut more easily than a short one.
- If the line of action of the force **passes through the axis**, the moment arm is zero and the torque is **zero** - pushing a door straight towards its hinges never opens it.



Door physics (textbook Figure 4.12): force at the handle, arm d from the hinge line.

> **Sign convention.** Anticlockwise torques are conventionally taken as **positive** and clockwise torques as negative. (Watch it: this is MCQ 1 of the exercise.)

Solved examples - torque

EXAMPLE 4.3 - OPENING A DOOR (BOOK, PAGE 114)

A force of **55 N** is applied to a door in three different ways, with moment arms (a) d = 0.80 m, (b) d = 0.60 m, and (c) d = 0 m. Find the torque in each case.

(a) tau = F x d = 55 x 0.80 = **44 Nm**

(b) tau = F x d = 55 x 0.60 = **33 Nm**

(c) tau = F x d = 55 x 0.00 = **0 Nm**

The force is the same in all three cases; only the moment arm changes, so the torque changes. In case (c) the line of action passes through the hinges, so no turning effect is produced at all.

ASSIGNMENT 4.3 - FORCE ON A BOTTLE OPENER (BOOK, PAGE 115) - SOLVED

A torque of 20 Nm is required to open a soda bottle. A boy applies a force perpendicularly at 0.1 m with a bottle opener. What force is required?

Given: tau = 20 Nm, d = 0.1 m. **Required:** F = ?

tau = F x d, so F = tau / d = 20 / 0.1 = **200 N**.

A longer opener would need less force: at 0.2 m the same 20 Nm needs only 100 N. This is the moment-arm principle at work.

Exam traps in torque questions. (1) Convert centimetres to metres before multiplying - 25 cm = 0.25 m. (2) A uniformly rotating fan has *zero net torque*, because its rotation is not changing. (3) Torque is not force: two different forces can produce the same torque if their moment arms differ.

4.6 Couple

> **Couple.** Two equal and opposite parallel forces acting along different lines of action on a body constitute a couple. A couple produces **rotation only**: the two forces cancel as forces (resultant force = 0, so no translation), but their turning effects add up - the resultant of a couple is a **pure moment**, not zero.

The shortest (perpendicular) distance between the two forces of a couple is called the **couple arm**.

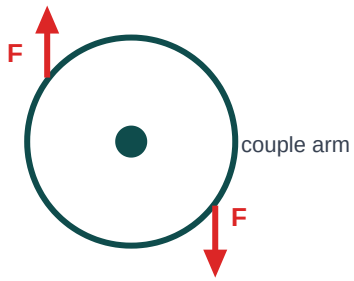
Torque of a couple = Force x Couple arm

It has the same value about every point of the body

Why the moment of a couple is the same about all points. Take moments about any point: the two equal forces F act at distances that always differ by exactly the couple arm, so the total moment works out to F x (couple arm) whatever point is chosen. The book lists this as a result you should be able to prove.

Everyday examples of a couple (from the book)

- Two hands turning a **steering wheel** - equal forces on opposite sides of the shaft.
- Forces on the two **bicycle pedals**.
- **Winding up** the spring of a toy car.
- Opening or closing the **cap of a bottle** with the fingers.
- Turning a **water tap**.
- Pressing two fingers on opposite edges of this physics book - however the forces are applied, the book turns.



A couple on a steering wheel: two equal, opposite forces, pure rotation.

4.7 Equilibrium

Equilibrium. The state of a body in which, under the action of several forces acting together, there is no change in its translational motion as well as its rotational motion. The study of bodies in equilibrium is called **statics**.

The two conditions of equilibrium

First condition: the vector sum of all the forces acting on the body is zero - sum F = 0. This prevents translation (no acceleration in any direction).

Second condition: the sum of all the torques acting on the body is zero - sum tau = 0. This prevents rotation.

Principle of moments

For a body in equilibrium, the sum of the clockwise moments about a pivot equals the sum of the anticlockwise moments about the same pivot.

WORKED BALANCE - TEXTBOOK FIGURE 4.17 (PAGE 118)

A rod is balanced on a central pivot. A weight W1 = 20 N hangs 2 m to the left of the pivot; weights W2 = 20 N at 1 m and W3 = 5 N at 4 m hang to the right.

Anticlockwise moment: tau1 = 20 x 2 = 40 Nm.

Clockwise moments: tau2 = 20 x 1 = 20 Nm; tau3 = 5 x 4 = 20 Nm; total = 40 Nm.

40 Nm - 20 Nm - 20 Nm = 0 Nm - the second condition is satisfied, so the rod balances. Note how one anticlockwise moment can be balanced by two clockwise moments added together.

Method for every balance problem. Choose the pivot, calculate each moment as force x distance from the pivot, put clockwise on one side and anticlockwise on the other, and set them equal. The seesaw numericals on pages 16-17 all use exactly this.

Types of equilibrium

Static equilibrium: the body is at rest under several forces and torques. Example: a book resting on a table - its weight mg is balanced by the normal reaction of the table.

Dynamic equilibrium: the body moves with uniform velocity under several forces. It is of two kinds: *translational* - a paratrooper falling at constant velocity; and *rotational* - a CD rotating in a player at constant angular velocity.

4.8 States of equilibrium and stability

Stability is a measure of how hard it is to displace an object from equilibrium - a stable object does not topple easily. The three states are decided by what happens to the centre of mass when the body is slightly displaced (textbook Figures 4.19 and 4.20, cone and ball-in-bowl tests):

State	When slightly displaced	Effect on centre of mass
Stable	The body returns to its original position when released (cone standing on its base; ball at the bottom of a bowl).	CM **rises**; the moment of the weight pulls the body back.
Unstable	The body moves further away and topples (cone balanced on its tip; ball on top of an upturned bowl).	CM **falls**; the moment of the weight increases the displacement.
Neutral	The body stays in its new position (cone lying on its side; ball on a flat surface).	CM **neither rises nor falls**; no moment acts either way.

Making an object more stable

- **Lower the centre of mass** - racing cars keep their CG/CM low and use wide tyres to reduce the chance of overturning.
- **Increase the area of support** - the bases of tall glasses and reading lamps are made heavy (which also lowers the CG) and large, so they stand firmly.
- Best of all, do **both**.

The toppling rule. A body topples when the vertical line through its CM falls outside its base. A wider base and a lower CM both mean the body must be tilted further before that line escapes - which is exactly why tall vehicles can overturn on sharp corners.

Formula sheet - the whole unit on one page

Quantity	Formula / statement
Horizontal component	Fx = F cos(theta)
Vertical component	Fy = F sin(theta)
Force from its components	F = square root of (Fx^2 + Fy^2)
Direction of the force	theta = $\tan^{-1}$(Fy / Fx)
Like parallel forces	Resultant = F1 + F2 (same direction)
Unlike parallel forces	Resultant = F1 - F2 (direction of the larger force)
Torque / moment of force	tau = F x d (SI unit: Nm)
Torque of a couple	tau = F x couple arm (same about all points)
First condition of equilibrium	Sum of all forces = 0
Second condition of equilibrium	Sum of all torques = 0
Principle of moments	Clockwise moments about pivot = anticlockwise moments about pivot
Weight (for numericals)	W = mg, with g = 10 m/s^2 at this level

Key points from the book (page 124)

- The SI unit of the moment of a force is the **newton metre (Nm)**.
- Moment of a force = force x perpendicular distance of the force from the pivot point.
- Resolution means splitting one vector into two or more components; addition is the reverse - combining vectors into a single resultant.
- Anticlockwise torque is positive; clockwise torque is negative.

> **Five mistakes that cost marks.** (1) Using the distance along the door instead of the *perpendicular* distance. (2) Forgetting to convert cm to m. (3) Calling a couple's resultant a force - its force is zero; only its moment survives. (4) Mixing up stable and neutral: check whether the CM rises (stable), falls (unstable) or stays level (neutral). (5) In balance problems, measuring a distance from the end of the rod instead of from the pivot.

Source: A Textbook of Physics, Grade IX, Khyber Pakhtunkhwa Textbook Board - Unit 4, pages 99-128.

Exercise MCQs - with answer key

The ten multiple-choice questions of the textbook exercise (page 126), each with the correct option and a one-line reason.

#	Question	Answer and reason
1	Conventionally, anticlockwise torque is taken as: A. negative B. positive C. parallel D. zero	**B - positive** (clockwise is negative).
2	A door needs a minimum torque of 80 Nm to open. Minimum distance of the handle from the hinges if the pull is not greater than 100 N: A. 0.6 m B. 1 m C. 0.4 m D. 0.8 m	**D - 0.8 m** (d = 80 / 100).
3	Two children balance on a seesaw. If one child leans inward toward the pivot, her side will: A. rise B. fall C. insufficient data D. neither	**A - rise** (her moment decreases).
4	A body in equilibrium must not be: A. at rest B. moving C. rotating D. accelerating	**D - accelerating** (it may move or rotate uniformly).
5	The torque in a uniformly rotating fan of blade length 0.5 m is: A. 0.5 Nm B. 2 Nm C. -0.5 Nm D. 0 Nm	**D - 0 Nm** (uniform rotation: net torque zero).
6	A force of 100 N applied perpendicularly at 0.5 m to turn the nut of a bus wheel gives a torque of: A. 500 Nm B. 50 Nm C. 5 Nm D. 0.005 Nm	**B - 50 Nm** (100 x 0.5).
7	The shortest distance between two couple forces is the: A. moment arm B. couple arm C. radius D. double moment	**B - couple arm**.
8	A girl pushes a door perpendicularly with 25 N at 0.6 m from the hinge; the torque is: A. 41.6 Nm B. 25.6 Nm C. 15 Nm D. 0 Nm	**C - 15 Nm** (25 x 0.6).
9	The angle at which the x and y components of a force are equal is: A. 0 degrees B. 30 degrees C. 45 degrees D. 60 degrees	**C - 45 degrees** (sin = cos).
10	CM differs from CG when we have a non-uniform: A. shape of object B. mass of object C. gravitational force D. none of these	**C - gravitational force**.

Solved numerical questions

All six numericals of the textbook exercise (page 128), solved step by step. Take g = 10 m/s^2 where a mass must be converted to weight.

NUMERICAL 1 - COMPONENTS OF A DOOR FORCE

To open a door, a force of 15 N is applied at 30 degrees to the horizontal. Find its horizontal and vertical components.

Fx = 15 cos 30 = 15 x 0.866 = **12.99 N** | Fy = 15 sin 30 = 15 x 0.5 = **7.5 N**

NUMERICAL 2 - TIGHTENING A BOLT

A bolt needs a torque of 40 Nm. You pull perpendicularly on the end of a 25-cm-long wrench. How much force must you exert?

d = 25 cm = 0.25 m. F = tau / d = 40 / 0.25 = **160 N**.

NUMERICAL 3 - BALANCING A SEESAW

Sana (mass 43 kg) sits 1.8 m from the centre of a seesaw. Faiz (mass 52 kg) wants to balance her. How far from the centre should Faiz sit?

Principle of moments: 52 x d = 43 x 1.8 (g cancels on both sides). d = 77.4 / 52 = **1.49 m** from the centre, on the opposite side.

NUMERICAL 4 - A THIRD KID JOINS THE SEESAW

Two kids of weights 300 N and 350 N sit at the ends of a 6 m seesaw pivoted at its centre. Where should a third kid of weight 250 N sit so the seesaw stays horizontal? (Ignore the weight of the seesaw.)

Each end is 3 m from the pivot. Moments: 300 x 3 = 900 Nm on one side; 350 x 3 = 1050 Nm on the other. Difference = 150 Nm, so the third kid must sit on the **300 N side**: 250 x d = 150, giving d = **0.6 m from the centre**.

Solved numerical questions (continued)

NUMERICAL 5 - TWO CHILDREN, ONE DOOR

Two children push on opposite sides of a door, both horizontally and perpendicular to it. One pushes with 20 N at 0.60 m from the hinges; the second pushes at 0.50 m. What force must the second child exert to keep the door from moving? (Friction is negligible.)

The door stays still when the moments balance: F x 0.50 = 20 x 0.60 = 12 Nm. F = 12 / 0.50 = **24 N**.

NUMERICAL 6 - MOMENT OF A CONSTRUCTION CRANE

A crane lifts building material of mass 1500 kg. Calculate the moment of force when the moment arm is 20 m, and again after the arm is raised so the moment arm becomes 12 m.

Weight of the load: W = mg = 1500 x 10 = 15,000 N.

At 20 m: moment = 15,000 x 20 = **300,000 Nm (3.0×10^5 Nm)**.

At 12 m: moment = 15,000 x 12 = **180,000 Nm (1.8×10^5 Nm)**.

Raising the arm shortens the moment arm, so the turning moment on the crane falls - one reason cranes lift loads close to the mast.

Answer summary

Numerical	Answer
1	Fx = 12.99 N, Fy = 7.5 N
2	F = 160 N
3	Faiz sits 1.49 m from the centre
4	Third kid sits 0.6 m from the centre, on the 300 N side
5	Second child must push with 24 N
6	300,000 Nm at 20 m; 180,000 Nm at 12 m

Conceptual questions - short answers

Brief model answers to the book's conceptual questions (page 127). Learn the idea, then write each answer in your own words.

1. Can a rectangular component be greater than the vector itself? No. Each component is F multiplied by cos(theta) or sin(theta), and neither can exceed 1, so a component is at most equal to F (when the other component is zero).

2. Why are door handles not put near the hinges? Torque = F x d. Near the hinges d is very small, so even a large force gives little torque and the door is hard to open. Handles are placed far from the hinges to maximise d.

3. Can a small force exert a greater torque than a larger force? Yes - if the small force acts at a much larger perpendicular distance from the axis. Torque depends on the product F x d, not on F alone.

4. Why is a long spanner better for a rusty nut? The longer spanner gives a larger moment arm, so the same force produces a larger torque to loosen the nut.

5. Gravity on a satellite always points to Earth's centre. Does it exert torque on the satellite? No. The force's line of action passes through the centre (the axis), so the moment arm is zero and the torque is zero.

6. Can a body fail to be in equilibrium even when the net force on it is zero? Yes - if the net torque is not zero. Examples: a couple applied to a steering wheel, and two equal opposite forces applied at different points of a door - the body rotates without translating.

7. Why do tightrope walkers carry a long rod? The rod lowers the walker's centre of mass and spreads the mass out, so small tilts produce restoring moments and balance is easier to keep.

8. Why does wearing high-heeled shoes sometimes cause lower back pain? High heels push the body's centre of mass forward over a very small base of support. The back muscles must keep exerting an extra moment to hold the body in equilibrium, and this constant strain produces lower back pain.

9. Why is it more difficult to lean backwards? Leaning back shifts the centre of mass behind the heels, so the line of action of the weight falls outside the base of support and its moment topples the body backwards. Leaning forward is easier because the toes extend the base in that direction.

10. Can a single force change both translational and rotational motion? Yes. A force whose line of action does not pass through the CM accelerates the CM (translation) and at the same time exerts a torque about the CM (rotation) - for example, striking a ball off-centre.

11. Two forces produce the same torque. Must they have the same magnitude? No. Equal torques only mean equal products F x d; a smaller force with a larger moment arm gives the same torque as a larger force with a smaller arm.

Comprehensive questions - where each answer lives

The book's long questions (pages 127-128) are all answered in the body of these notes. Use this checklist when revising for the board exam:

#	Question	See page
1	What are force diagrams? Define like and unlike parallel forces with examples.	Page 3
2	Explain the addition of forces in connection with the head-to-tail rule.	Page 4
3	Define moment of a force, give its mathematical description, and elaborate the factors on which it depends.	Pages 9-10
4	What is resolution of forces? Explain with an example how a force is resolved into rectangular components.	Pages 5-6
5	What is a couple? Explain with examples.	Page 11
6	Define equilibrium, explain its types, and state the two conditions of equilibrium.	Pages 12-13
7	State and explain the principle of moments with examples.	Page 12
8	What are centre of mass and centre of gravity? How can CM/CG be determined? Is there any difference between them?	Pages 7-8
9	Explain the stability of objects with reference to the position of the centre of mass.	Page 13

Lab work of the unit (book, page 120)

- Verify the principle of moments using a metre rod balanced on a wedge.
- Find the weight of an unknown object using the principle of moments: balance a known weight against the unknown on a metre rod, then (unknown weight) x (its distance from the pivot) = (known weight) x (its distance from the pivot).

> **Final revision routine.** Day 1: pages 3-8 (concepts). Day 2: pages 9-13 (torque, couple, equilibrium). Day 3: page 14 from memory, then pages 15-18 without looking at the answers first. Any question you miss goes back into your notebook in your own handwriting.

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